

MS 6015: Multivariate Statistical Methods for Business

Logistic regression

Outline

Parameter Estimation and Interpretation

Model Diagnostics

Omnibus: Likelihood Ratio Test

Wald's Test

Logistic regression as a classifier

Sensitivity, Specificity and Precision

Optimal cut-off probability

Variable Selection in Logistic Regression

Forward LR (Likelihood Ratio)

Forward Selection Wald

Example: Universal Bank

Regression analysis progression

01

Simple linear

$$y = \beta_0 + \beta_1 x + \varepsilon$$

- Single predictor variable
- Linear relationship between x and y

02

Multiple linear

$$y = \beta_0 + \beta_1 x_1 + \cdots + \beta_p x_p + \varepsilon$$

- Multiple predictor variables
- Each $\hat{\beta}_j$ captures partial effect

03

Multicollinearity

High correlation among predictors

- Inflates standard errors
- Unstable parameter estimates

04

Categorical

Dummy variable encoding

- $k - 1$ dummies for k categories
- One group as base level

Motivation

- ▶ Classification problems are an important category of problems in analytics in which the response variable (Y) takes a discrete value.
- ▶ There are several techniques used for solving classification problems such as logistic regression, classification trees, discriminant analysis, neural networks, and support vector machines.

Logistic Regression

- ▶ Logistic regression (LR) is a statistical technique for finding existence of a relationship between a qualitative (discrete) dependent variable (or outcome variable) and several independent variables (aka explanatory variables or predictors).
- ▶ In LR, the primary objective is to find the conditional probability of occurrence of an event (class probability) given the values of several independent variables. LR is one of the most popular tools used for solving classification problems.

Binomial and Multinomial Logistic Regression

- ▶ Binomial (or binary) logistic regression is a model in which the dependent variable is dichotomous.
- ▶ In its simplest setting, the response variable Y is restricted to two values.
 - ▶ For example, Y may be recorded as “male” or “female” or “employed” and “not employed.”
- ▶ In multinomial logistic regression model, the dependent variable can take more than two values.
- ▶ The independent variables may be of any type.

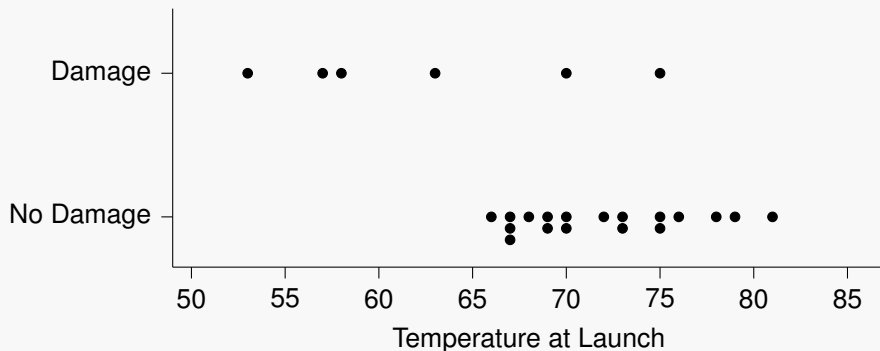
Example: Space Shuttle Challenger Data

- ▶ Space shuttle orbiter Challenger (Mission STS-51-L) was the 25th shuttle launched by NASA on January 28, 1986
 - ▶ The Challenger crashed 73 seconds into its flight due to the erosion of O-rings which were part of the solid rocket boosters of the shuttle.
 - ▶ Before the launch, the engineers at NASA were concerned about the outside temperature which was very low.
- ▶ Data shows the O-ring erosion and the launch temperature of the previous shuttle launches, where
 - ▶ 'damage to O-ring = 1' implies there was a damage to O-ring
 - ▶ 'damage to O-ring = 0' implies there was no damage to O-ring during that launch.

Example: Space Shuttle Challenger Data

- ▶ Space shuttle orbiter Challenger (Mission STS-51-L) was the 25th shuttle launched by NASA on January 28, 1986
 - ▶ The Challenger crashed 73 seconds into its flight due to the erosion of O-rings which were part of the solid rocket boosters of the shuttle.
 - ▶ Before the launch, the engineers at NASA were concerned about the outside temperature which was very low.
- ▶ Data shows the O-ring erosion and the launch temperature of the previous shuttle launches, where
 - ▶ 'damage to O-ring = 1' implies there was a damage to O-ring
 - ▶ 'damage to O-ring = 0' implies there was no damage to O-ring during that launch.
- ▶ Goal: To develop a logistic regression model to predict the probability of erosion of O-ring based on the launch temperature.

Challenger launch temperature vs damage data



Drawbacks of Single Predictor Model

Let the response Y be either 0 or 1. If we were to model the probability of 1 with a single predictor linear model, we would write

$$E(Y|X) = \beta_0 + \beta_1 X$$

Drawbacks

- ▶ The predicted values of the response Y could become greater than 1 or less than 0 because the linear expression for its expected value is unbounded.
- ▶ One of the assumptions of a regression analysis is that the variance of Y is constant across all values of the predictor variable X .

The Fix: The Logit Model

- ▶ Instead of modeling the probability p directly with a linear model, we first consider the odds ratio

$$odds = \frac{p}{1 - p}$$

which is the ratio of the probability of 1 to the probability of 0.

- ▶ Unlike probability, the odds ratio can be greater than 1.

Logit

- ▶ In logistic regression for a binary variable, we model the natural log of the odds ratio, which is called $\text{logit}(p)$. Thus

$$\text{logit}(p) = \ln \left(\frac{p}{1-p} \right)$$

Logit as a function of p

- ▶ The logit is a function of the probability p . In the simplest model, we assume that the logit graphs as a straight line in the predictor variable X so

$$\text{logit}(p) = \ln \left(\frac{p}{1-p} \right) = \beta_0 + \beta_1 X$$

- ▶ Because it is easier for most people to think in terms of probabilities, we can convert from the logit or log odds to the probability p .

Logistic Function

- By first exponentiating

$$\ln\left(\frac{p}{1-p}\right) = \beta_0 + \beta_1 X$$

we get

$$\theta = \frac{p}{1-p} = \exp(\beta_0 + \beta_1 X)$$

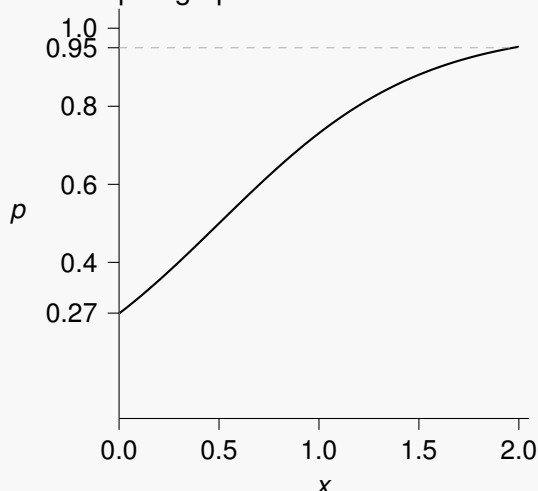
Solving for θ we get

$$p = \frac{\exp(\beta_0 + \beta_1 X)}{1 + \exp(\beta_0 + \beta_1 X)}$$

which describes a logistic curve.

Logistic Curve

The relation between p and the predictor X is not linear but has an S-shaped graph



Parameter Estimation

- ▶ One of the major assumptions of multiple linear regression model is that the residuals follow a normal distribution (or approximate normal distribution).
- ▶ However, the residuals in logistic regression will not follow a normal distribution and thus we cannot use the method of ordinary least squares (OLS) to estimate the regression parameters.
- ▶ Regression parameters in the case of logistic regression are estimated using the Maximum Likelihood Estimator (MLE).

Limitations of MLE

- ▶ Maximum likelihood estimator may not be unique or may not exist.
- ▶ Closed form solution may not exist for many cases, one may have to use iterative procedure to estimate the parameter values.

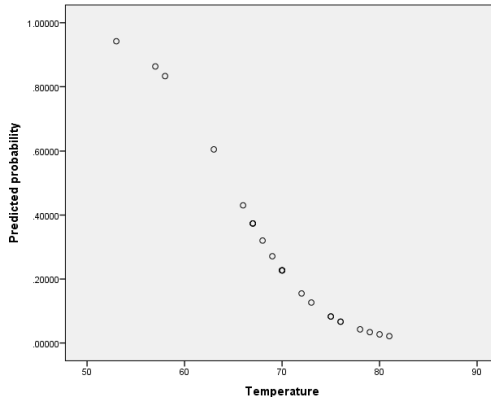
└ Parameter Estimation and Interpretation

Variables in the Equation

	B	S.E.	Wald	df	Sig.	Exp(B)
Step 1 ^a Temperature	-.236	.107	4.832	1	.028	.790
Constant	15.297	7.329	4.357	1	.037	4398676.183

a. Variable(s) entered on step 1: Temperature.

Graph



Interpretation of Parameters

- Interpretation of logistic regression parameters is not as simple as in the case of linear regression. Consider the logit function $\ln \left(\frac{P(Y_i=1)}{1-P(Y_i=1)} \right) = \beta_0 + \beta_1 X_i$

Interpretation of Parameters

- Interpretation of logistic regression parameters is not as simple as in the case of linear regression. Consider the logit

$$\text{function } \ln \left(\frac{P(Y_i=1)}{1-P(Y_i=1)} \right) = \beta_0 + \beta_1 X_i$$

$$\ln \left(\frac{P(Y_i=1|X_i=0)}{1-P(Y_i=1|X_i=0)} \right) = \beta_0$$

Interpretation of Parameters

- Interpretation of logistic regression parameters is not as simple as in the case of linear regression. Consider the logit

$$\text{function } \ln \left(\frac{P(Y_i=1)}{1-P(Y_i=1)} \right) = \beta_0 + \beta_1 X_i$$

$$\ln \left(\frac{P(Y_i=1|X_i=0)}{1-P(Y_i=1|X_i=0)} \right) = \beta_0$$

$$\ln \left(\frac{P(Y_i=1|X_i=1)}{1-P(Y_i=1|X_i=1)} \right) = \beta_0 + \beta_1$$

Interpretation of Parameters

- Interpretation of logistic regression parameters is not as simple as in the case of linear regression. Consider the logit function $\ln \left(\frac{P(Y_i=1)}{1-P(Y_i=1)} \right) = \beta_0 + \beta_1 X_i$

$$\ln \left(\frac{P(Y_i=1|X_i=0)}{1-P(Y_i=1|X_i=0)} \right) = \beta_0$$

$$\ln \left(\frac{P(Y_i=1|X_i=1)}{1-P(Y_i=1|X_i=1)} \right) = \beta_0 + \beta_1$$

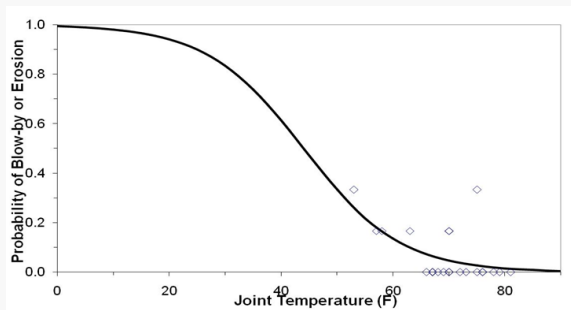
$$\beta_1 = \ln \left(\frac{\left(\frac{P(Y_i=1|X_i=1)}{1-P(Y_i=1|X_i=1)} \right)}{\left(\frac{P(Y_i=1|X_i=0)}{1-P(Y_i=1|X_i=0)} \right)} \right)$$

That is, β_1 is the change in the log-natural of the ratio of odds. Alternatively, $\exp(\beta_1)$ or is the change in the odds ratio for unit change in the feature value

Interpretation of the Regression Coefficient

- ▶ When β_1 is positive, $P(Y = 1)$ increases as the value of the predictor X_i increases.
- ▶ When β_1 is negative, $P(Y = 1)$ decreases as the value of the predictor X_i increases.

Challenger data



The probability of damage decreases as the launch temperature increases.

Model Diagnostics

- ▶ Omnibus test: Omnibus tests are generic class of tests in statistics which check whether the explained variance in the model is significantly higher than the unexplained variation.
- ▶ Wald's test: Wald's test is used for checking statistical significance of individual predictor variables (equivalent to t-test in MLR model).
- ▶ Hosmer-Lemeshow test: Hosmer-Lemeshow (H-L) is a chi-square goodness of fit test used for checking the goodness of logistic regression model.
- ▶ Pseudo R^2 : It is not possible to calculate R^2 as in the case of continuous dependent variable in a logistic regression model. However, many pseudo R^2 values are used which compare the intercept-only model to the model with independent variables.

- └ Model Diagnostics

- └ Omnibus: Likelihood Ratio Test

Outline

Parameter Estimation and Interpretation

Model Diagnostics

Omnibus: Likelihood Ratio Test

Wald's Test

Logistic regression as a classifier

Sensitivity, Specificity and Precision

Optimal cut-off probability

Variable Selection in Logistic Regression

Forward LR (Likelihood Ratio)

Forward Selection Wald

Example: Universal Bank

Omnibus: Likelihood Ratio Test

- ▶ In logistic regression model, likelihood ratio test is used as the test for checking the statistical significance of the overall model.
- ▶ The null and alternative hypothesis for a logistic regression model with k independent variables are given by
 - ▶ $H_0 : \beta_1 = \beta_2 = \dots = \beta_k = 0$
 - ▶ $H_1 : \text{not all } \beta \text{ are zero}$
- ▶ It is evident that the likelihood ratio test is statistically significant (since the p -value is 0.003).
 - ▶ That is, the predictor variable, launch temperature, is statistically significant.

Likelihood Function

- The log likelihood function for binary logistic regression model is given by

$$LL = \sum_{i=1}^n Y_i \ln[\pi(Z)] + \sum_{i=1}^n (1 - Y_i) \ln[(1 - \pi(Z))]$$

where $P(Y = 1) = \pi(Z)$

Analysis of Deviance

- ▶ Let N be the total number of observations in the data and let N_0 be the number of 0s (or negatives) and N_1 be the number of 1s (or positives) in the data set. Then, in the absence of any logistic regression model (i.e., model without any independent variable), the estimates of $P(Y = 1)$ and $P(Y = 0)$ are given by

$$P(Y = 1) = \frac{N_1}{N}, P(Y = 0) = \frac{N_0}{N}$$

- ▶ Null deviance: $-2 \times LL_0 = -2[N_0 \cdot \ln(\frac{N_0}{N}) + N_1 \cdot \ln(\frac{N_1}{N})]$
- ▶ Residual deviance:

$$-2 \times LL_M = -2 \sum_{i=1}^n Y_i \ln[\pi(Z)] + \sum_{i=1}^n (1 - Y_i) \ln[(1 - \pi(Z))]$$

- ▶ $-2LL_0 - (-2LL_M) = -2\ln(\frac{L_0}{L_M})$.

This follows an approximate chi-square distribution. It calculates reduction in deviance which is similar to reduction in SSE after adding a variable in case of MLR model.

- └ Model Diagnostics
 - └ Wald's Test

Outline

Parameter Estimation and Interpretation

Model Diagnostics

Omnibus: Likelihood Ratio Test

Wald's Test

Logistic regression as a classifier

Sensitivity, Specificity and Precision

Optimal cut-off probability

Variable Selection in Logistic Regression

Forward LR (Likelihood Ratio)

Forward Selection Wald

Example: Universal Bank

Wald's Test

- ▶ Wald's test is used for checking statistical significance of individual predictor variables (equivalent to t -test in MLR model). The null and alternative hypotheses for Wald's test are:
 - ▶ $H_0 : \beta_i = 0$
 - ▶ $H_1 : \beta_i \neq 0$
- ▶ Wald's test statistic is given by

$$W = \left[\frac{\hat{\beta}_i}{se(\hat{\beta}_i)} \right]^2$$

Wald's test is a chi-square test with degrees of freedom = 1.

- ▶ Since the p -value is less than 0.05, we reject the null hypothesis.
 - ▶ That is, the variable 'launch temperature' has a statistically significant relationship on O-ring failure.

Pseudo R^2

- ▶ Cox and Snell R^2 :

$$R^2 = 1 - \left\{ \frac{L(\text{Intercept} - \text{only model})}{L(\text{FullModel})} \right\}^{2/N}$$

- ▶ Nagelkerke's R^2 :

$$R^2 = \frac{1 - \left\{ \frac{L(\text{Intercept} - \text{only model})}{L(\text{FullModel})} \right\}^{2/N}}{1 - L(\text{Intercept} - \text{only Model})^{2/N}}$$

Classification table

- ▶ The primary objective of logistic regression is to solve classification problems based on the predicted probability values.
- ▶ The output from a logistic regression model is the class probability $P(Y = 1)$.
- ▶ Based on the value of $P(Y = 1)$, the decision maker has to classify the observation as belonging to either class 1 (positive) or class 0 (negative).
- ▶ Once a logistic regression function has been established, and using training sets for each of the two populations, we can proceed to classify.

Confusion Matrix

		Predicted	
		Negative	Positive
Actual	Negative	TN True Negative	FP False Positive
	Positive	FN False Negative	TP True Positive

Classification Table/ Confusion Matrix

		Cutoff:	0.50	
	Predicted:			
Actual:	# No	# Yes	Total	
# No	17	0	17	
# Yes	3	4	7	
Total	20	4	24	
Percent correct =	87.5%		TPR=	57.1%

	Predicted:		
Actual:	% No	% Yes	Total
% No	71%	0%	71%
% Yes	13%	17%	29%
Total	83%	17%	100%
	True negative rate =	100.0%	

Accuracy paradox

- ▶ The accuracy paradox in classification problem states that a model with higher overall accuracy may not be a better model.
- ▶ For example, in the case of Challenger data, when the classification cut-off probability is 0.5, the overall accuracy is 87.5 and when the classification cut-off probability is 0.2, the overall accuracy is only 62.5.
- ▶ However, given the context, we need higher accuracy in predicting positive classes than negative classes
- ▶ The accuracy in predicting positive classes when classification cutoff is 0.5 is 57.1, whereas the corresponding accuracy for classification cut-off probability of 0.2 is 85.7.
- ▶ In this case, classification cut-off 0.2 is better than 0.5 although the overall classification accuracy is lesser for classification cut-off probability 0.2.

- └ Logistic regression as a classifier
- └ Sensitivity, Specificity and Precision

Outline

Parameter Estimation and Interpretation

Model Diagnostics

Omnibus: Likelihood Ratio Test

Wald's Test

Logistic regression as a classifier

Sensitivity, Specificity and Precision

Optimal cut-off probability

Variable Selection in Logistic Regression

Forward LR (Likelihood Ratio)

Forward Selection Wald

Example: Universal Bank

- └ Logistic regression as a classifier
- └ Sensitivity, Specificity and Precision

Sensitivity, Specificity and Precision

- ▶ In Logistic Regression, the model performance is often measured using concepts such as sensitivity, specificity, precision, and F -score.
- ▶ The ability of the model to correctly classify positives and negatives is called sensitivity and specificity respectively.
- ▶ The terminologies sensitivity and specificity originated in medical diagnostics.

- └ Logistic regression as a classifier
- └ Sensitivity, Specificity and Precision

Sensitivity (TPR, Recall)

- ▶ The ability of the model to correctly classify positives and negatives are called sensitivity and specificity, respectively.
- ▶ In general
Sensitivity = $P(\text{model classifies } Y_i \text{ as positive} \mid Y_i \text{ is positive})$
Sensitivity is calculated using the following equation:

$$\text{Sensitivity} = \frac{TP}{TP + FN}$$

where True Positive (TP) is the number of positives correctly classified as positives by the model and False Negative (FN) is positives misclassified as negative by the model. Sensitivity is also called as recall.

- └ Logistic regression as a classifier
- └ Sensitivity, Specificity and Precision

Specificity (TNR)

- ▶ Specificity is the ability to correctly classify the test as negative.
- ▶ In general:
Specificity = $P(\text{model classifies } Y_i \text{ as negative} \mid Y_i \text{ is negative})$
Specificity can be calculated using the following equation:

$$\text{Specificity} = \frac{TN}{TN + FP}$$

where True Negative (TN) is number of the negatives correctly classified as negatives by the model and False Positive (FP) is number of negatives misclassified as positives by the model.

- └ Logistic regression as a classifier
- └ Sensitivity, Specificity and Precision

Confusion Matrix

		Predicted			
		Negative	Positive		
Actual	Negative	TN True Negative	FP False Positive	Specificity = TNR $TN / (TN + FP)$	
	Positive	FN False Negative	TP True Positive	Sensitivity = TPR $TP / (TP + FN)$	

- └ Logistic regression as a classifier
- └ Sensitivity, Specificity and Precision

Precision

- ▶ There is a trade-off between sensitivity and specificity in most classification problems.
- ▶ The decision maker has to consider the tradeoff between sensitivity and specificity to arrive at an optimal cut-off probability.
- ▶ Precision measures the accuracy of positives classified by the model.

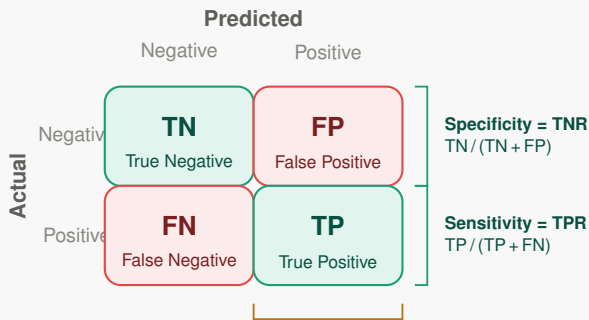
$$Precision = \frac{TP}{TP + FP}$$

- ▶ F Score (F Measure) is another measure used in binary logistic regression that combines both precision and recall and is given by:

$$F - score = \frac{2 \times Precision \times Recall}{Precision + Recall}$$

- └ Logistic regression as a classifier
- └ Sensitivity, Specificity and Precision

Confusion matrix with key metrics



$$\text{Precision (PPV)} = TP / (TP + FP)$$

F_1 -Score (harmonic mean of Precision & Recall)

$$F_1 = \frac{2 \cdot TP}{2 \cdot TP + FP + FN} = 2 \cdot \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

- └ Logistic regression as a classifier
- └ Sensitivity, Specificity and Precision

Classification Metrics Overview

- ▶ **Accuracy:** Measures overall correctness.
- ▶ **Precision:** Indicates accuracy of positive predictions.
- ▶ **Recall (Sensitivity):** Measures the proportion of actual positives correctly identified.
- ▶ **Specificity:** Measures the proportion of actual negatives correctly identified.

Metrics	Values
Sensitivity (Recall)	$\frac{TP}{TP+FN}$
Specificity	$\frac{TN}{TN+FP}$
Accuracy	$\frac{TP+TN}{TP+TN+FP+FN}$
Precision	$\frac{TP}{TP+FP}$

- └ Logistic regression as a classifier
- └ Sensitivity, Specificity and Precision

When does each metric matter?

Metric	Formula	Answers	Use when ...
Sensitivity (TPR / Recall)	$TP / (TP + FN)$	Of all actual positives, how many were detected?	Missing a positive is costly — cancer screening, disease diagnosis
Specificity (TNR)	$TN / (TN + FP)$	Of all actual negatives, how many correctly rejected?	False alarms costly — ruling out disease, avoiding unnecessary treatment
Precision (PPV)	$TP / (TP + FP)$	Of all predicted positives, how many truly positive?	False positives costly — spam filter, legal document retrieval
F₁-Score	$2 TP / (2 TP + FP + FN)$	Single number balancing precision and recall	Class imbalance makes accuracy misleading — fraud detection

- └ Logistic regression as a classifier
- └ Sensitivity, Specificity and Precision

Concordant and Discordant Pairs

The concepts of concordant and discordant pairs are measures used to assess the logistic regression model performance.

- ▶ **Discordant Pairs.** A pair of positive and negative observations for which the model has no cut-off probability to classify both of them correctly are called discordant pairs.
- ▶ **Concordant Pairs.** A pair of positive and negative observations for which the model has a cut-off probability to classify both of them correctly are called concordant pairs.

A logistic regression model with high proportion of concordant pairs is preferred

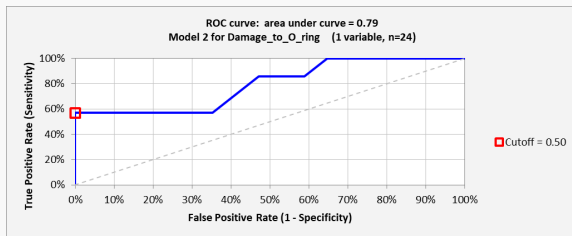
- └ Logistic regression as a classifier
- └ Sensitivity, Specificity and Precision

ROC Curve and AUC

- ▶ Receiver operating characteristic (ROC) curve can be used to understand the overall worth of a logistic regression model
- ▶ ROC curve is a plot between sensitivity (true positive rate) in the vertical axis and $1 - \text{specificity}$ (false positive rate) in the horizontal axis.
- ▶ **ROC Curve:** Plots TPR against FPR at different thresholds.
- ▶ **AUC (Area Under Curve):** Measures the area under the ROC curve. The AUC is the proportion of the concordant pairs in the data. Model with higher AUC is preferred and AUC is frequently used for model selection class).
- ▶ In case of imbalanced datasets, the AUC may be very high (greater than 0.9); however, either sensitivity or specificity values may be poor. In such cases, Lift is preferred for model selection.

- └ Logistic regression as a classifier
- └ Sensitivity, Specificity and Precision

ROC for Challenger data



- └ Logistic regression as a classifier
- └ Sensitivity, Specificity and Precision

Interpretation of AUC

For challenger crash data, the AUC is 0.794. The area under the ROC curve can be interpreted as follows:

1. If we use the logistic regression model, then there will be 79.4% concordant pairs and 20.6% discordant pairs.
2. For a randomly selected pair of positive and negative observations, probability of correctly classifying them is 0.794.
3. For a randomly selected positive and negative observations, the following relationship is valid:
$$P[P(\text{Positive}) > P(\text{Negative})] = 0.794$$

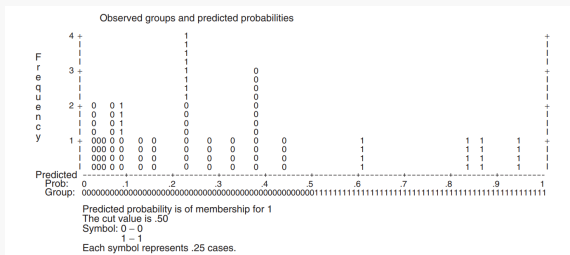
Optimal cut-off probability

While using logistic regression model, one of the decisions that a data scientist has to make is to choose the right classification cut-off probability (P_c). The overall accuracy, sensitivity and specificity will depend on the chosen cut-off probability. The following three methods are used for selecting the cut-off probability.

1. Classification plot
2. Youden's Index
3. Cost based approach

Classification Plot for Selection of Cut-off Probability

- Classification plot is a plot between the probability on horizontal axis and predicted probability for each of the observations using LR model.

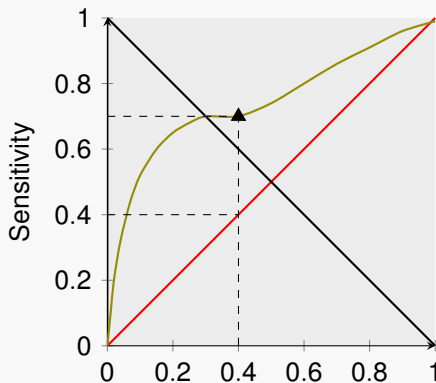


Although, classification plot may help us to understand how sensitivity and specificity are likely to change when we change the classification cut-off probability, arriving at an optimal cut-off is difficult using classification plot.

Youden's Index for Optimal Cut-Off Probability

- ▶ Youden's Index is a classification cut-off probability, P_c , for which the following function is maximized (also known as J statistic):

$$J - \text{statistic} = \text{Max}_p[\text{Sensitivity}(p) + \text{Specificity}(p) - 1]$$



Cost-based cut-off probability

- ▶ We use Youden's Index only when both sensitivity and specificity are equally important.
- ▶ When sensitivity and specificity are not equally important then we use cost-based approach for cut-off probability.
- ▶ In cost-based approach, we assign penalty cost for misclassification of positives and negatives. Assume that cost of misclassifying negative (0) as positive (1) is C_{01} and cost of misclassifying positive (1) as negative (0) is C_{10} as shown below:

Observed	Classified	
	0	1
0		C_{01}
1	C_{10}	

- ▶ The optimal cut-off probability is the one which minimizes the total penalty cost and is given by
$$\text{Min}_p[C_{01}P_{01} + C_{10}P_{10}]$$

Variable Selection in Logistic Regression

Motivation

Automatically select variables so that only **statistically significant** predictors (at significance level α) are included in the logistic regression model.

Two main approaches discussed here:

1. **Forward LR** (Likelihood Ratio)
2. **Forward Selection Wald**

- └ Variable Selection in Logistic Regression
 - └ Forward LR (Likelihood Ratio)

Outline

Parameter Estimation and Interpretation

Model Diagnostics

Omnibus: Likelihood Ratio Test
Wald's Test

Logistic regression as a classifier

Sensitivity, Specificity and Precision

Optimal cut-off probability

Variable Selection in Logistic Regression

Forward LR (Likelihood Ratio)
Forward Selection Wald

Example: Universal Bank

- └ Variable Selection in Logistic Regression
 - └ Forward LR (Likelihood Ratio)

Forward LR -Overview

- ▶ At each step, **one variable is added** to the model.
- ▶ Selection is based on the **Likelihood Ratio (LR) test** (omnibus test).
- ▶ A variable is added only if its p -value is $\leq \alpha$.

Uses the statistic $-2LL$ (negative twice the log-likelihood) to measure model fit improvement.

- └ Variable Selection in Logistic Regression
 - └ Forward LR (Likelihood Ratio)

Forward LR - Step 1

Step 1

Start with **no variables** in the model.

Set $i = 0$.

- └ Variable Selection in Logistic Regression
 - └ Forward LR (Likelihood Ratio)

Forward LR - Step 2

Step 2

For each independent variable, calculate the difference between $-2LL_i$ and $-2LL_{i+1}$:

- ▶ When $i = 0$: compare $-2LL_0$ (no variable) vs. $-2LL_1$ (one variable).
- ▶ Add the variable with the **highest difference** in $-2LL$, provided:
 - ▶ The p -value from the LR test is **statistically significant** at level α .

- └ Variable Selection in Logistic Regression
 - └ Forward LR (Likelihood Ratio)

Forward LR - Step 3

Step 3

Repeat Step 2 until one of the following:

- ▶ All variables have been exhausted, or
- ▶ The change in $-2LL$ is not significant (i.e., p -value after adding a new variable $> \alpha$).

- └ Variable Selection in Logistic Regression
 - └ Forward Selection Wald

Outline

Parameter Estimation and Interpretation

Model Diagnostics

Omnibus: Likelihood Ratio Test
Wald's Test

Logistic regression as a classifier

Sensitivity, Specificity and Precision

Optimal cut-off probability

Variable Selection in Logistic Regression

Forward LR (Likelihood Ratio)

Forward Selection Wald

Example: Universal Bank

Forward Selection Wald - Overview

- ▶ Variables are entered based on **Wald's test** rather than the LR test.
- ▶ At each step the variable with the **smallest p -value** (or equivalently, largest Wald statistic) is added, provided $p \leq \alpha$.

Forward Selection Wald - Step 1

Step 1

Assume the data has n explanatory variables.

- ▶ Develop a **univariate** logistic regression model for each variable.
- ▶ Calculate the p -value of all variables using Wald's test.
- ▶ Add the variable with the **smallest p -value**, if it is less than α .

Forward Selection Wald - Step 2

Step 2

Add a new variable with:

- ▶ The **smallest p -value**, or
- ▶ The **largest Wald statistic value**,

... if the p -value (based on Wald's test) is less than the significance level α .

- └ Variable Selection in Logistic Regression

- └ Forward Selection Wald

Forward Selection Wald - Step 3

Step 3

Repeat the procedure until the p -value becomes **greater than** α .

Forward LR vs. Forward Selection Wald

Forward LR

- ▶ Uses **Likelihood Ratio** test
- ▶ Compares $-2LL$ between nested models
- ▶ Adds variable with highest $\Delta(-2LL)$
- ▶ Stops when LR p -value $> \alpha$

Forward Wald

- ▶ Uses **Wald's test**
- ▶ Fits univariate models first
- ▶ Adds variable with smallest Wald p -value
- ▶ Stops when Wald p -value $> \alpha$

Both methods include only **statistically significant** variables at level α .

Background

- ▶ A relatively young bank is growing rapidly in terms of overall customer acquisition.
- ▶ Majority of these are Liability customers with varying sizes of relationship with the bank.
- ▶ The customer base of Asset customers is quite small, and the bank WANTS to grow this base rapidly to bring in more loan business.
- ▶ Specifically, it wants to explore ways of converting its liability customers to Personal Loan customers.
- ▶ A campaign the bank ran for liability customers last year showed a healthy conversion rate of over 9% successes.
- ▶ This has encouraged the Retail Marketing department to devise smarter campaigns with better target marketing.

Analytics Objectives

- ▶ While designing a new campaign, can we model the previous campaign's customer behavior to analyze what combination of parameters make a customer more likely to accept a personal loan?
- ▶ There are several special products / facilities the bank offers like CD(certificate of Deposit) and security accounts, online services, credit cards, etc.
- ▶ Can we spot any association among these for finding cross-selling opportunities?

Data Description

ID	Customer ID
Age	Customer's age in completed years
Experience	#years of professional experience
Income	Annual income of the customer (\$000)
ZIPCode	Home Address ZIP code.
Family	Family size of the customer
CCAvg	Avg. spending on credit cards per month (\$000)
Education	Education Level. 1: Undergrad; 2: Graduate; 3: Advanced/Professional
Mortgage	Value of house mortgage if any. (\$000)
Personal Loan	Did this customer accept the personal loan offered in the last campaign?
Securities Account	Does the customer have a securities account with the bank?
CD Account	Does the customer have a certificate of deposit (CD) account with the bank?
Online	Does the customer use internet banking facilities?
CreditCard	Does the customer use a credit card issued by UniversalBank?

Personal Loan Offer

- ▶ Outcome variable: accept bank loan (0/1)
- ▶ Predictors: Demographic info,(Age, Income etc) and info about their bank relationship(mortgage, securities, account etc.), Customer response to personal loan campaign.
- ▶ Among 5000 customers, only 480 accepted loan offer.
- ▶ Goal: Find characteristics of customers who are most likely to accept loan offerings in future.

Data Preprocessing

- Create 0/1 dummy variables for categorical predictors

$$EducProf = \begin{cases} 1 & \text{if education is } Professional \\ 0 & \text{otherwise} \end{cases}$$

$$EducGrad = \begin{cases} 1 & \text{if education is at } Graduate \text{ level} \\ 0 & \text{otherwise} \end{cases}$$

$$Securities = \begin{cases} 1 & \text{if customer has securities account in bank} \\ 0 & \text{otherwise} \end{cases}$$

$$CD = \begin{cases} 1 & \text{if customer has CD account in bank} \\ 0 & \text{otherwise} \end{cases}$$

$$Online = \begin{cases} 1 & \text{if customer uses online banking} \\ 0 & \text{otherwise} \end{cases}$$

$$CreditCard = \begin{cases} 1 & \text{if customer holds Universal Bank credit card} \\ 0 & \text{otherwise} \end{cases}$$

Single Predictor Model

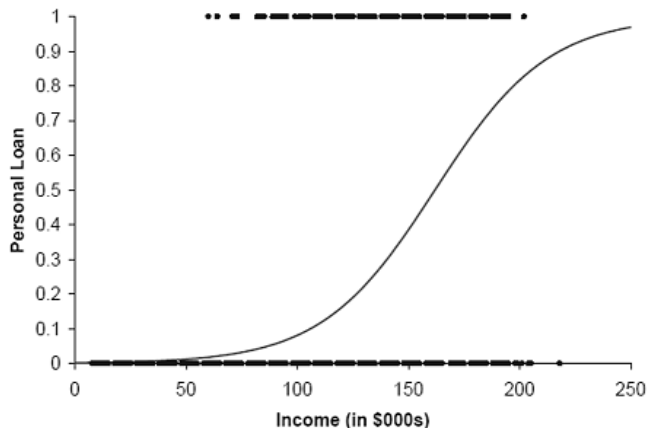
Modeling loan acceptance on income (x)

$$P(\text{Personal Loan} = \text{Yes} | \text{Income} = x) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x)}}$$

Fitted coefficients (more later): $b_0 = -6.127$, $b_1 = 0.037$

└ Example: Universal Bank

Seeing the Relationship



Last step - classify

Model produces an estimated probability of being a “1”

- ▶ Convert to a classification by establishing cutoff level
- ▶ If estimated prob. $>$ cutoff, classify as “1”

Ways to Determine Cutoff

- ▶ 0.50 is popular initial choice
- ▶ Additional considerations
 - ▶ Maximize classification accuracy
 - ▶ Maximize sensitivity (subject to min. level of specificity)
 - ▶ Minimize false positives (subject to max. false negative rate)
 - ▶ Minimize expected cost of misclassification (need to specify costs)

Example: Continued

- ▶ Estimates of β 's are derived through an iterative process called maximum likelihood estimation
- ▶ Let's include all 12 predictors in the model now
- ▶ Output gives coefficients for the logit, as well as odds for the individual terms

└ Example: Universal Bank

Regression Model

The Regression Model

Input variables	Coefficient	Std. Error	p-value	Odds
Constant term	-13.20165825	2.46772742	0.00000009	*
Age	-0.04453737	0.09096102	0.62439483	0.95643985
Experience	0.05657264	0.09005365	0.5298661	1.05820346
Income	0.0657607	0.00422134	0	1.06797111
Family	0.57155931	0.10119002	0.00000002	1.77102649
CCAvg	0.18724874	0.06153848	0.00234395	1.20592725
Mortgage	0.00175308	0.00080375	0.02917421	1.00175464
Securities Account	-0.85484785	0.41863668	0.04115349	0.42534789
CD Account	3.46900773	0.44893095	0	32.10486984
Online	-0.84355801	0.22832377	0.00022026	0.43017724
CreditCard	-0.96406376	0.28254223	0.00064463	0.38134006
EducGrad	4.58909273	0.38708162	0	98.40509796
EducProf	4.52272701	0.38425466	0	92.08635712

Figure 10.3: Logistic regression coefficient table for personal loan acceptance as a function of 12 predictors.

Estimated Equation for Logit

Based on training data

$$\begin{aligned}\text{logit} = & -13.201 - 0.045\textit{Age} + 0.057\textit{Experience} + 0.066\textit{Income} + 0.572\textit{Family} \\ & + 0.18724874\textit{CCAvg} + 0.002\textit{Mortgage} - 0.855\textit{Securities} + 3.469\textit{CD} \\ & - 0.844\textit{Online} - 0.964\textit{Credit Card} + 4.589\textit{EducGrad} + 4.523\textit{EducProf}\end{aligned}$$

Equation for Odds

$$\begin{aligned} \text{odds}(\textit{Personal Loan} = \text{Yes}) = & e^{-13.201} (0.956)^{\textit{Age}} (1.058)^{\textit{Experience}} (1.068)^{\textit{Income}} \\ & \cdot (1.771)^{\textit{Family}} (1.206)^{\textit{CCAvg}} (1.002)^{\textit{Mortgage}} \\ & \cdot (0.425)^{\textit{Securities}} (32.105)^{\textit{CD}} (0.430)^{\textit{Online}} \\ & \cdot (0.381)^{\textit{CreditCard}} (98.405)^{\textit{EducGrad}} (92.086)^{\textit{EducProf}} \end{aligned}$$

Converting to probability

$$p = \frac{\textit{Odds}}{1 + \textit{Odds}}$$

Interpreting Odds, Probability

- ▶ For predictive classification, we typically use probability with a cutoff value
- ▶ For explanatory purposes, odds have a useful interpretation:
 - ▶ If we increase x_1 by one unit, holding $x_2, x_3, \dots, x_q$ constant, then b_1 is the factor by which the odds of belonging to class 1 increase

Loan Example: Evaluating Classification Performance

Performance measures: Confusion matrix and % of misclassifications

Summary

- ▶ Logistic regression is similar to linear regression, except that it is used with a categorical response
- ▶ It can be used for explanatory tasks (=profiling) or predictive tasks (=classification)
- ▶ The predictors are related to the response Y via a nonlinear function called the logit
- ▶ As in linear regression, reducing predictors can be done via variable selection
- ▶ Logistic regression can be generalized to more than two classes